Practice workbook · Distance travelled by integrating velocity

Finding total distance by integrating velocity, splitting at times of rest and adding sizes.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-29

1. Question 3 · part b · 4 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part b only.

3 A particle P moves in a straight line. At time t seconds, the velocity, v m/s, of P is given by

$$v = t^2 - 4t + 7$$

(a) Find the acceleration of P, in m/s^2, when $t = 3$

(2)

(b) Find the distance, in m, that P travels in the interval $0 \leqslant t \leqslant 6$

(4)

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1. Question 3 · part b · 4 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part b only.



Question 3 continued

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(Total for Question 3 is 6 marks)

1. Question 3 · part b · 4 marks · Working and answer

Working and answer

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2. Question 1 · part c · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

1 A particle P moves in a straight line. At time t seconds, the velocity, v m/s, of P is given by

$$v = 3t^2 - 9t + 7$$

(a) Show that P never comes to rest.

(2)

(b) Find the acceleration of P, in m/s^2, when $t = 4$

(2)

Using algebra

(c) find the distance, in m, that P travels in the interval $0 \leqslant t \leqslant 5$

(3)

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P 7 2 8 6 5 A 0 4 3 6

2. Question 1 · part c · 3 marks

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Further practice · Attempt part c only.

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Question 1 continued

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(Total for Question 1 is 7 marks)

2. Question 1 · part c · 3 marks · Working and answer

Working and answer

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3. Question 4 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

4 A particle P moves along the x-axis. At time t seconds ($t \geqslant 0$) the acceleration, a m/s^2, of P is given by $a = 6t - 12$

When $t = 0$, P is at rest at the origin.

(a) Find the velocity of P when $t = 2$

(3)

At time T seconds, $T > 0$, P is instantaneously at rest.

(b) Find the value of T.

(2)

(c) Find the distance travelled by P in the first 8 seconds of its motion.

(3)

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3. Question 4 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 8 marks)

3. Question 4 · part c · 3 marks · Working and answer

Working and answer

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4. Question 4 · part b · 8 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

4 A particle P is moving along the x-axis. At time t seconds, $t \geqslant 0$, the velocity, v m/s, of P is given by $v = 2t^2 - 16t + 30$

(a) Find the acceleration, in m/s^2, of P when $t = 5$

(2)

P comes to instantaneous rest at the points M and N at times t_1 seconds and t_2 seconds where $t_2 > t_1$

(b) Find the exact distance MN

(8)

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4. Question 4 · part b · 8 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 10 marks)

4. Question 4 · part b · 8 marks · Working and answer

Working and answer

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5. Question 4 · part c · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

4 A particle P is moving along a straight line. At time t seconds $(t \geqslant 0)$, its velocity, v m/s, is given by $v = 4t^2 - 6t + 5$

(a) Find the minimum speed of P

(3)

The acceleration of P at time T seconds is $18\ \text{m/s}^2$

(b) Find the value of T

(2)

When $t = 0$, P is at the point X, and when $t = 3$, P is at the point Y

(c) Find the distance XY

(4)

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5. Question 4 · part c · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 9 marks)

5. Question 4 · part c · 4 marks · Working and answer

Working and answer

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6. Question 5 · part c · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

5 A particle P is moving along the x-axis.

At time t seconds, $t \geqslant 0$, the velocity, v m/s, of P is given by

$$v = 2t^2 - 19t + 35$$

(a) Find the acceleration of P when $t = 5$

(2)

The particle comes to instantaneous rest at the points A and B at times t_1 seconds and t_2 seconds respectively, where $t_1 < t_2$

(b) Find the value of t_1 and the value of t_2

(2)

(c) Use calculus to find the distance AB

(3)

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6. Question 5 · part c · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 7 marks)

6. Question 5 · part c · 3 marks · Working and answer

Working and answer

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7. Question 5 · part c · 5 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

5 A particle P is moving along a straight line.

At time t seconds $(t \geqslant 0)$, the velocity, v m/s, of P is given by

$$v = 3t^2 - 16t + 5$$

(a) Find the values of t when P is instantaneously at rest.

(3)

At time t seconds the acceleration of P is a m/s^2

(b) Find the range of values of t for which $a > 0$

(2)

(c) Find the distance, in m, that P travels in the interval $1 \leqslant t \leqslant 4$

(5)

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7. Question 5 · part c · 5 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 10 marks)

7. Question 5 · part c · 5 marks · Working and answer

Working and answer

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8. Question 3 · part c · 4 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

3 A particle P is moving along a straight line, from a fixed origin O

At time t seconds $(t \geqslant 0)$, the velocity, v m/s, of P is given by

$$v = 2t^2 - 19t + 35$$

At time t seconds the acceleration of P is a m/s^2

(a) Find the value of t for which $a = 0$

(3)

P is instantaneously at rest at time T_1 seconds and at time T_2 seconds where $T_2 > T_1$

(b) Find the value of T_1 and the value of T_2

(3)

(c) Find the exact distance, in metres, that P travels between the times T_1 and T_2
Show your working clearly.

(4)

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8. Question 3 · part c · 4 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 3 continued

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(Total for Question 3 is 10 marks)

8. Question 3 · part c · 4 marks · Working and answer

Working and answer

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